

Mode 1 — Baseline-BFP

DP16 (8×4) → DP8 (8×4) → block exponents

Step 1 — DP16 (8×4)

$$\text{DP16 } (8 \times 4) = \sum_{i=0}^{15} a_i b_i$$

Step 2 — DP8 (8×4)

$$\text{DP16 } (8 \times 4) = \underbrace{\sum_{i=0}^7 a_i b_i}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=8}^{15} a_i b_i}_{\text{DP8 } (8 \times 4)}$$

Step 3 — Block exponents (one exponent block per DP8 (8×4))

$$\left(\underbrace{\sum_{i=0}^7 a_i b_i}_{\text{DP8 } (8 \times 4)}, E_0 \right), \left(\underbrace{\sum_{i=8}^{15} a_i b_i}_{\text{DP8 } (8 \times 4)}, E_1 \right) \longrightarrow (M, E)$$

$E_0 = E_{a,0} + E_b$, $E_1 = E_{a,1} + E_b$; $E = \max(E_0, E_1)$, $\delta_k = E - E_k$; truncating shifts, on block sums only; $\text{DP16 } (8 \times 4) = M \cdot 2^E$.

Step 4 — Result

$$M = 2^{-\delta_0} \sum_{i=0}^7 a_i b_i + 2^{-\delta_1} \sum_{i=8}^{15} a_i b_i, \quad E = \max(E_0, E_1)$$